The Silent Golden Station

The Harmonic Fingerprint of the Cube–Octahedron Sweep Finds Icosaxes, and Only Icosaxes

Theorem: along the sweep from the octahedron to the cube, the cubic harmonic of degree 4 vanishes at one solid only, Icosaxes at $\mu = \varphi$, where every degree with no icosahedral harmonic falls silent; degree 8 vanishes there and at one bronze station

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The test came from DICTO's widening of the search to harmonics, waves and resonance (The DICTO Concordance, Part 5) and his call to run it. Computation, proofs and drafting with Claude (Anthropic).

Abstract. Give every corner of a solid seen from its centre a unit weight on the sphere and expand in spherical harmonics: the strength Q_l of each degree l is the solid's harmonic fingerprint, the bond-orientational order of Steinhardt, Nelson and Ronchetti [1], which physicists use to tell the cubic, hexagonal and icosahedral order of atoms apart. We apply it to #90's sweep $P_\mu = C \cap \mu C^*$ of the cube and its polar octahedron, from the octahedron ($\mu = 1$) to the cube ($\mu = 3$). **Main theorem.** (A) Q_4 vanishes at exactly one point of the sweep, $\mu = \varphi$, Icosaxes; there the corner condition is $a^4 - 3a^2 + 1 = (a^2 + a - 1)(a^2 - a - 1) = 0$ with $a = \mu - 1$. (B) At Icosaxes Q_l vanishes exactly for $l = 2, 4, 8, 14$, the even degrees with no icosahedral harmonic: its 24 corners are the two icosahedra of the pyritohedral pair (#90), so the one cubic solid on the sweep sounds icosahedral. (C) Q_8 vanishes at exactly two points, $\mu = \varphi$ and $\mu = (\sqrt{13} - 1)/2$, the reciprocals of the golden and the bronze ratio. (D) On the second dual sweep, the cuboctahedron cut by the rhombic dodecahedron, Q_4 vanishes once, at a point with no golden number, $1 + \sqrt{((6\sqrt{30} - 31)/17)}$. The proofs are short polynomial identities; every value is checked numerically against the classical fingerprints of the cubic, hexagonal and body-centred packings and the icosahedron.

A solid can be listened to: its corners, seen from the centre, are notes on a sphere, and each degree of spherical harmonic is a tone. Cubic solids ring in the fourth tone, icosahedral ones never do. Sliding from the octahedron to the cube, the fourth tone falls silent once, at exactly the golden station, Icosaxes, because its corners are two icosahedra. The sound finds the golden point with no geometry put in.

1. The fingerprint

For a finite set of directions $u_1, \ldots, u_N$ put $q_{lm} = (1/N) \Sigma Y_{lm}(u_i)$ and $Q_l = (4\pi/(2l + 1) \Sigma_m |q_{lm}|^2)^{1/2}$ [1]. Q_l does not change under rotation; $\hat{W}_l$ is the normalised third-order invariant built with Wigner 3j symbols [12]. Since Frank's observation that liquids favour icosahedral order [6] and the polytetrahedral picture of Nelson and Spaepen [7], these numbers have become the standard local test of order in simulations of crystals, liquids and glasses [8, 9], with

known limits [10]. For a set with the full cube–octahedron symmetry every q_{lm} lies in the space of cube-invariant harmonics of degree l, which for l = 4, 6, 8 is one-dimensional, spanned (Meyer [11]) by $K_4 = x^4 + y^4 + z^4 - (3/5)r^4$, K_6 and $K_8 = \Sigma x^8 - 14\Sigma x^6y^2 + 35\Sigma x^4y^4$ (sums over the distinct terms; harmonic, checked). Hence **$Q_l = 0$ exactly when $\Sigma K_l(u_i) = 0$**, which turns each zero into a polynomial equation.

Corner set	Q_4	Q_6	Q_8	Q_{10}	Q_{12}
Cuboctahedron (cubic close packing)	0.191	0.575	0.404	0.013	0.600
Anticuboctahedron (hexagonal close packing)	0.097	0.485	0.317	0.010	0.565
Body-centred shell (8 + 6)	0.036	0.511	0.429	0.195	0.405
Icosahedron	0	0.663	0	0.363	0.585
Dodecahedron	0	0.368	0	0.560	0.020
Cube (= tetrahedron)	0.509	0.628	0.213	0.650	0.415
Octahedron	0.764	0.354	0.718	0.411	0.696

The first four rows are the classical values [1] and serve as the check of the code. The table shows two limits of the method: the even tones do not hear a point reflection, so the tetrahedron sounds exactly like the cube (its corners and their mirror images are the cube's), and the rhombic dodecahedron's corner directions are the body-centred shell's. The third-order invariant $\hat{W}_6$ does separate the dual pair dodecahedron (+0.170) and icosahedron (−0.170): the same tone, opposite sign, corners against faces.

2. Main theorem

> **Main theorem.** Let $P_\mu = C \cap \mu C^*$, $C = [-1, 1]^3$, $C^* = \{|x| + |y| + |z| \le 1\}$, $1 < \mu < 3$ (#90 [2]).
> (A) $Q_4(P_\mu) = 0$ if and only if $\mu = \varphi$ (Icosaxes).
> (B) At $\mu = \varphi$, $Q_l = 0$ for l = 2, 4, 8, 14 and $Q_l \neq 0$ for the other even $l \le 16$.
> (C) $Q_8(P_\mu) = 0$ if and only if $\mu = \varphi$ or $\mu = (\sqrt{13} - 1)/2$.
> (D) On the sweep $CO \cap \mu RD$ from the rhombic dodecahedron ($\mu = 1$) to the cuboctahedron ($\mu = 2$), with $CO = \{|x| \le 1, |x| + |y| + |z| \le 2\}$ and RD its polar, $Q_4 = 0$ at exactly one point, $\mu_0 = 1 + a_0$, $17a_0^4 + 62a_0^2 - 7 = 0$, so $a_0^2 = (6\sqrt{30} - 31)/17$ and $\mu_0 \approx 1.3311$.
> (E) On the tetrahedron's own dual sweep $T \cap \mu(-T)$ (truncated tetrahedra, octahedron at $\mu = 1$), $Q_4 = 0$ exactly at the root $\mu \approx 0.5538$ of $9\mu^4 - 52\mu^3 + 30\mu^2 - 4\mu + 1 = 0$ and at its mirror $1/\mu$; neither is the golden cut ($\mu = (3 + 2\sqrt{5})/11$ or $(\varphi + 2)/(3\varphi + 2)$) nor the regular truncated tetrahedron ($\mu = 3/5$).

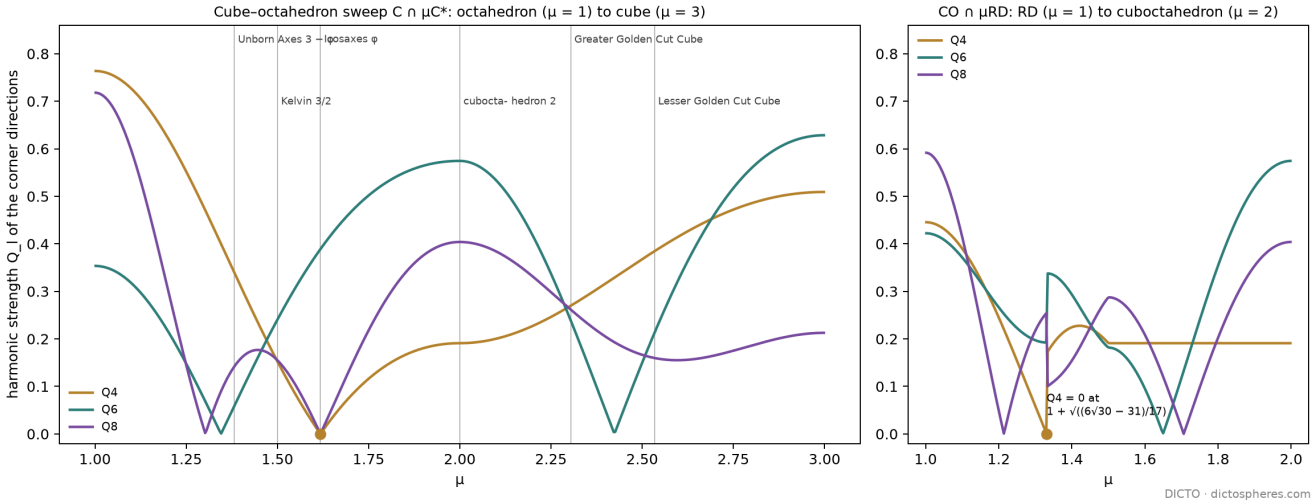


Figure 1. The fingerprint along the two sweeps. Left: from the octahedron to the cube, with Unborn Axes, Kelvin's cell, Icosaxes, the cuboctahedron and the two Golden Cut Cubes marked; Q_4 (gold) and Q_8 (violet) touch zero together only at Icosaxes, and Q_8 once more at the bronze station $\mu = (\sqrt{13} - 1)/2 \approx 1.303$. Right: from the rhombic dodecahedron to the cuboctahedron; Q_4 touches zero once, at a point with no golden number.

Proof. For $1 < \mu \le 2$ the corners of P_μ are the 24 points $(\pm 1, \pm a, 0)$ in all orders, $a = \mu - 1$; for $2 \le \mu < 3$ the 24 points $(\pm 1, \pm 1, \pm b)$, $b = \mu - 2$ (#90). (A) For (1, a, 0), $K_4/r^4 = (1 + a^4)/(1 + a^2)^2 - 3/5$, which vanishes when $2a^4 - 6a^2 + 2 = 0$, that is $a^4 - 3a^2 + 1 = (a^2 + a - 1)(a^2 - a - 1) = 0$; in (0, 1) only $a = 1/\varphi$, $\mu = \varphi$. For (1, 1, b), $(2 + b^4)/(2 + b^2)^2 = 3/5$ gives $b^4 - 6b^2 - 1 = 0$, whose positive root $b^2 = 3 + \sqrt{10}$ is above 1. (B) At $a = 1/\varphi$ the corners are the cyclic arrangements of $(0, \pm 1, \pm 1/\varphi)$ in both orders, the vertices of two icosahedra (the compound of #90, Figure 1 (c) [2]). Each icosahedron alone has $q_{lm} = 0$ in every degree that carries no icosahedral harmonic, and the even degrees that carry one are 0, 6, 10, 12, 16, 18, … (the degrees $6p + 10q$ [3]); so the mean over both vanishes at $l = 2, 4, 8, 14$. The non-vanishing at $l = 6, 10, 12, 16$ is the computed value (0.389, 0.313, 0.357, 0.403). (C) With K_8, the sum over the corners (1, a, 0) is $6(a^2 + a - 1)(a^2 - a - 1)(a^2 + 3a - 1)(a^2 - 3a - 1)$; in (0, 1) the roots are $a = 1/\varphi$ and $a = (\sqrt{13} - 3)/2 = 1/\beta_3$, $\beta_3 = (3 + \sqrt{13})/2$ the bronze ratio. On (2, 3) the sum over the corners (1, 1, b) is $3(b^8 - 28b^6 + 70b^4 - 28b^2 + 9)$, which has no root in (0, 1); Q_8 stays above 0.155. (D) For $1 < \mu < 4/3$ the corners are the 24 points $(\pm 1, \pm a, \pm a)$ in all orders and the 8 points $(\pm \mu/2, \pm \mu/2, \pm \mu/2)$ (computed face census: 6 squares and 12 hexagons). The eight give $K_4/r^4 = -4/15$ each, so $Q_4 = 0$ when $24((1 + 2a^4)/(1 + 2a^2)^2 - 3/5) = 32/15$, i.e. $17a^4 + 62a^2 - 7 = 0$. On (4/3, 2) no zero was found on a grid of 2000 points; from $\mu = 3/2$ on, Q_4 stays at the cuboctahedron's 0.1909 to four decimals. ∎

3. What the sound hears

> *The cubic tone and the eighth tone fall silent together only where the corners become two icosahedra, and the eighth tone alone hears one more of the metallic ratios, the bronze.*

The fourth-degree condition $a^4 - 3a^2 + 1 = 0$ is the golden equation twice, for a and for $-a$; nothing else on the sweep can silence the cubic tone. The eighth-degree condition adds the bronze pair $a^2 \pm 3a - 1$, the next metallic ratio after the golden one with the same form $x^2 = nx + 1$ ($n = 1$ golden, $n = 3$ bronze); the silver ratio ($n = 2$), which carries the Golden Cut Cubes' second field in #90, does not appear. The bronze station $\mu = (\sqrt{13} - 1)/2$ is not one of the solids named so far; its corners are $(1, 1/\beta_3, 0)$ and it sits between the octahedron and

Unborn Axes. Q_6 also vanishes twice on the sweep, at $a^2 = (17 - \sqrt{273})/4$ ($\mu \approx 1.345$) and at $\mu \approx 2.424$, near Unborn Axes and the truncated cube but at neither.

The second dual sweep, from the rhombic dodecahedron to the cuboctahedron, has no such station: its one zero of Q_4 involves $\sqrt{30}$ and no golden number. So the icosahedral sound belongs to the first pair, the cube and the octahedron, where #48 and #90 found Icosaxes and Unborn Axes in the first place.

4. Networks: every node by its own sound

The fingerprint was made for networks: physicists compute it at every atom from its nearest neighbours to tell the local order. The same works on our networks. On the stacking ABCBACB, repeated, the twelve neighbours of each ball give $Q_4 = 0.191$ at every straight ball and 0.097 at every twisted one, so the fingerprint reads off the c/h word of the Hexagonality Law papers ball by ball, with no knowledge of the layers (checked on all seven letters). In a network of pieces it does the same for any node: the set of directions to its neighbours, or to the corners of the pieces around it, has a fingerprint that does not depend on how the node is turned, so equal nodes are found wherever they sit, and icosahedral nodes ($Q_4 = Q_8 = 0$) stand out among cubic ones.

5. Prior art, and the status of each result

Known, credited. The invariants Q_l, $\hat{W}_l$ and their classical values for the cubic, hexagonal and body-centred shells and the icosahedron: Steinhardt, Nelson and Ronchetti [1]. The degrees of icosahedral harmonics: Klein's invariants of degrees 6, 10, 15, and [3]. The sweep and its stations: #90 [2], #48 [4]; Icosaxes as the hull of two icosahedra: #90 [2]. The metallic ratios are classical [13]. The idea of hearing the solids is Kepler's [14]; the icosahedral cluster of Mackay [15] is the physical icosahedral shell.

Proved here. (A)–(E). We did not find in the sources checked a fingerprint of the cube–octahedron truncation family, nor the golden and bronze zeros. The classical uses fingerprint the neighbour shells of atoms; here the same numbers are read along a one-parameter family of solids, where they pick out its golden station.

Open. The fingerprint of the golden nests and of the OG shells against the icosahedral clusters of quasicrystals; the odd degrees ($l = 3, 7, 9$), which hear the tetrahedron's mirror image; the reason Q_4 is constant from $\mu = 3/2$ to 2 on the second sweep; the sweep from the icosidodecahedron to the rhombic triacontahedron.

Data and reproducibility

studies/2026-11-04-harmonic-fingerprint: fingerprint.py (all Q_l, $\hat{W}_6$, the sweep, data/fingerprint.log, .json), fig_sweeps.py (Figure 1); studies/2026-11-04-co-rd-sweep: co_rd_sweep.py (face census and Q_4 on the second sweep). Run on dicto-big-brother with Python 3, numpy, scipy and sympy (exact polynomials for K_4, K_6, K_8).

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